

1. The problem: a signal too small to see

1. Set up your breadboard with the RPi providing 5V as you have done in previous labs.
2. Place the PIN diode on the breadboard in reverse bias, the same way you oriented the photo-diode last week. Use a 1 k Ω resistor as the load resistor, connecting it between the diode and ground.
3. Connect the oscilloscope probe across the load resistor (probe tip to the side connected to the diode, ground clip to the ground side).
4. Turn on the oscilloscope and push the auto-set button if needed.
5. Try shining your phone flashlight at the PIN diode. Can you see any change in the voltage on the oscilloscope?

We did not see a change in the voltage on the oscilloscope.

6. If you can see a signal at all, how large is it? Record the approximate voltage.

Think about it: The BPW34 datasheet says this diode produces about 75 μ A of photocurrent at 1000 lux (bright indoor light). With your 1 k Ω load resistor, what voltage would that produce across RL? (Use Ohm's law: $V = I \times R$.) What if you used a 1 M Ω load resistor instead? We will come back to this question.

For detecting radiation, the flashes of light we need to measure are much fainter than a phone flashlight, and they happen in millionths of a second. Even with amplification, this signal will be challenging. We need a smarter approach.

2. Amplifying the signal with an op-amp

2a.

1. Take the LM358 op-amp chip (A diagram of the op-amp pins is included at the end of this lab) and place it across the center gap on the breadboard so that the pins straddle the gap. The chip has a small notch at the "top" with pin 1 on the left.
2. Connect the chip's power: pin 8 (V+ - top right) goes to the +5V rail, and pin 4 (GND) goes to the ground rail.
3. Build the mid-supply reference: connect two 10 k Ω resistors in series between the +5V rail and ground. Connect the midpoint (where the two resistors meet) to pin 3 (the non-inverting input, marked +).
- 3.1. If you have trouble getting a clean signal - also connect a 1 μ F capacitor from this midpoint to ground - this smooths out any noise on the reference voltage (see the modified Figure 2 at the end of the lab).
4. Connect the PIN diode so that the anode goes to pin 2 - the inverting (-) input.
5. Connect a 1 M Ω feedback resistor between pin 1 (output) and pin 2 (inverting input).
6. Connect the oscilloscope probe to pin 1 (output) with the ground clip to the ground rail.

7. Before shining any light on the diode, what voltage do you see at the output? It should be near 2.5V.

8. Now shine your phone flashlight at the PIN diode. What happens to the output voltage?

When you shine your phone flashlight at the PIN diode, the output voltage will decrease from the 2.5V baseline.

Think about it: The BPW34 produces about 75 μA at 1000 lux. With a 1 M Ω feedback resistor, how much output swing would that require? How does that compare to the 2.5V of headroom you have below mid-supply?

You have just discovered an important limitation: the op-amp's output can only swing within its power supply rails. When the required output swing exceeds what the supply allows, the signal clips. We can address this in two ways: reduce the feedback resistor (less gain) or increase the supply voltage (more headroom). Let's try increasing the supply.

2b.

1. Connect the boost converter to the 5V RPi supply. Use the voltmeter to dial the output up to approximately 10V. Verify this voltage with the meter before connecting anything else.

2. Reconnect the op-amp power: pin 8 (V+) now goes to the +10V boost converter output, and pin 4 (GND) stays on the ground rail.

3. Update the voltage divider: connect two 10 k Ω resistors between the +10V output and ground (replacing the connection to the 5V rail). The midpoint is now at 5V and still connects to pin 3.

3.1. The 1 μF bypass capacitor may become necessary now – why?

The capacitor reduces the noise introduced by the op-amp power.

4. First, try with the 1 M Ω feedback resistor still in place. Shine your flashlight. Does it still clip? Yes (It likely will – even 5V of headroom is only enough for 5 μA with a 1 M Ω resistor.)

5. Now swap the feedback resistor to 10 k Ω . With this resistor, clipping occurs at 500 μA – well above what the BPW34 produces in normal lighting.

6. Try shining your phone flashlight at the PIN diode. You should now see the output voltage change when you illuminate the diode. Try waving your hand over the diode, or covering and uncovering it.

7. Record the output voltage with the flashlight on and off. Calculate the photocurrent using $I = (V_{\text{ref}} - V_{\text{out}}) / R_f$, where V_{ref} is your mid-supply voltage ($\approx 5\text{V}$).

The output voltage with no light is 5V, when the flashlight is on, it decreases to about 3.7V. The photocurrent for darkness is $(5-5)/(1 \times 10^6) = 0$. The photocurrent for flashlight is $(5-3.7)/(1 \times 10^6) = 0.0000013$

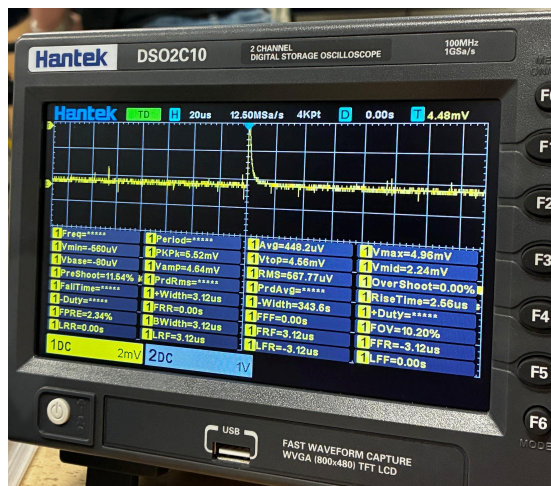
Think about it: You have now seen the same signal problem solved two different ways.

In Part 1, you used a load resistor, and $V = I \times R$ gave the same conversion factor as the TIA. Why is the op-amp circuit better? (Hint: think about what the op-amp does to the voltage across the diode – the op-amp holds the reference voltage at a fixed value, which keeps the diode at a constant bias point regardless of the signal current. This is

not true if we just use a load resistor.)

4. Building the detector circuit

1. Set up your breadboard with the RPi providing 5V.
2. Connect the boost converter to the 5V supply. Use the voltmeter to dial the output up to approximately 29V. Verify this voltage with the meter before connecting anything else.
3. The detector diagram (Figure 4) shows the pins you will need. You will use two connections: CATHODE (bias voltage) and ANODE (signal output).
4. Connect the boost converter output (+29V) to the CATHODE pin. Place a 0.22 μF capacitor (please pay attention to the capacitor orientation – it matters!) between the cathode connection and ground – this is a filtering capacitor to smooth out noise from the boost converter. Place a small resistor (around 50 Ω), in series between the boost converter output and the cathode, before the capacitor. This resistor and capacitor together form an RC low-pass filter.
5. Connect a load resistor (try 200 Ω to start, you can also try up to 1 k Ω) between the ANODE pin and the GND rail.
6. Connect the oscilloscope probe across the load resistor: probe tip to the anode side, ground clip to the ground rail.
7. Turn on the oscilloscope. Adjust the trigger level so that it is just below the baseline.
8. Try bringing a radioactive source near the detector. You should see a change in the pulse rate.
9. Take a screenshot or picture of your oscilloscope output and include it here.



5. Comparing signals and thinking about noise

1. Compare the pulses you see on your oscilloscope to the ones from the demonstration with the clean power supply. What differences do you notice?

The pulses on our oscilloscope have a lot more noise and our peak is sharper than the peak on the graph from the demonstration.

2. You will likely see more noise – random fluctuations – on your signal compared to the clean supply. Where is this noise coming from?

This noise is coming from the fact that our power supply isn't clean because our power source contains inconsistencies like voltage and frequency variations. This is most likely caused by a less than ideal circuit.

3. The RC filter on the bias line (the resistor and capacitor you placed between the boost converter and the cathode) is trying to block the high-frequency switching noise from the boost converter while letting the steady 29V through.

4. What could you change about the filter to block more noise? What tradeoff might that involve?

To block more noise, we could increase the filter's order, adjust the cutoff frequency, or have better grounding. However, this might involve a slower response from the system or signal distortion.

5. Try changing the load resistor value (for example, switch from 200 Ω to 1 k Ω). How does this change the pulse shape? What about the noise level?

Increasing the resistor value would cause the circuit to respond slower, causing the peak on the graph to look more rounded rather than sharp. The noise level would go down.

6. Comment on the pulses you are seeing. Are they what you expected? Think about what is happening physically: a gamma ray hits the crystal, light is produced, the SiPM converts it to current, and that current flows through your load resistor.

Yes this is what we expected. There's a lot of small pulses, but it can also be affected by noise.

7. Look at the pulse heights. Are all the pulses the same height, or do they vary? If you wanted to build an energy spectrum from these pulses, you would need to digitize each pulse height. Think about what ADC parameters (sampling rate, number of bits) would matter most for accurately recording the pulse heights.

The pulses vary in height. The number of bits is very important to accurately measure pulse height.